

Week 3 — Mapping the Local Market

Goal: Turn last week's job *titles* into a target list of real *openings* at real companies you could commute to. By the end of this week you'll know your local engineering economy better than most adults in town.

Part 1: Research

A. Company census

Find **every company within your commuting range** that does mechanical or electrical engineering work: manufacturers, machine shops, electronics firms, utilities, HVAC/controls companies, defense contractors, medical device makers, automation/robotics integrators, engineering consultancies.

Head start: `resources/north-texas-starter-map.md` is a researched, sourced map of the McKinney/Collin County area — Raytheon and Prysmian in town, the TI Sherman fab up the road, and more. Start from it, verify each entry is still current, and then find what it misses: your census needs at least 5 companies that aren't on that map (the small shops are where the self-made opportunities are).

Target: **at least 12 companies**. For each:

- Name, location, rough size (LinkedIn shows employee count)
- What they actually make or do (one sentence)
- Whether their careers page mentions tuition reimbursement, apprenticeships, or "education assistance" (this matters for Week 4)
- Whether they mention **interns or co-op students** anywhere on the careers page — flag these companies; they matter later (see `resources/internships-and-coops.md`)

Where to look:

- Google Maps: search "manufacturing," "machine shop," "electronics manufacturer," "engineering" around your area and just scroll
- Your county or state's Economic Development site — most publish a "major employers" list
- Local chamber of commerce member directory
- LinkedIn company search filtered by location + industry
- Drive past any industrial park you already know of and write down the names on the buildings — seriously, this works, and half those companies never post on job boards

B. Openings list

From those companies (plus Indeed/LinkedIn), collect **15 real, currently-open, entry-level positions** you could plausibly apply to. For each: title, company, pay (if listed), requirements, link, and application deadline if any.

Part 2: Essay (600–900 words)

Prompt: *“Three companies I could see myself in.”*

Requirements:

- Choose 3 companies from your census and go deep: what they make, who their customers are, what kinds of engineers they employ, what entry roles they have open or have posted recently.
- For each, answer: *what would I be able to learn here that I couldn't learn at a random job?*
- Rank them. Say what would make you pick #1 over #2.
- Note anything you found about tuition help at any of them, even a rumor to verify later.

Done when

- ☐ 12+ company census complete
- ☐ 15 real openings collected (links saved — postings expire, screenshot the good ones)
- ☐ Essay written
- ☐ Walkthrough conversation done

Keep the openings list safe. Weeks 6 and 8 are built directly on it.